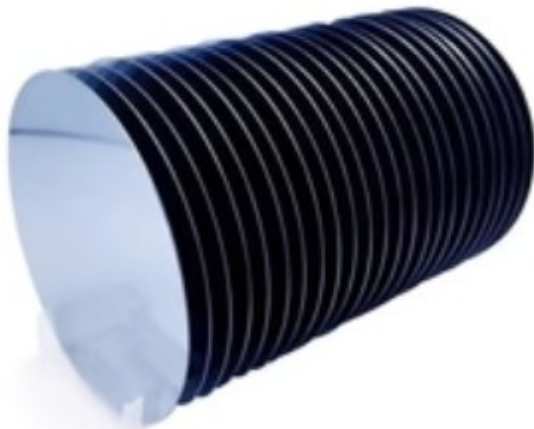


Diodes and Transistors

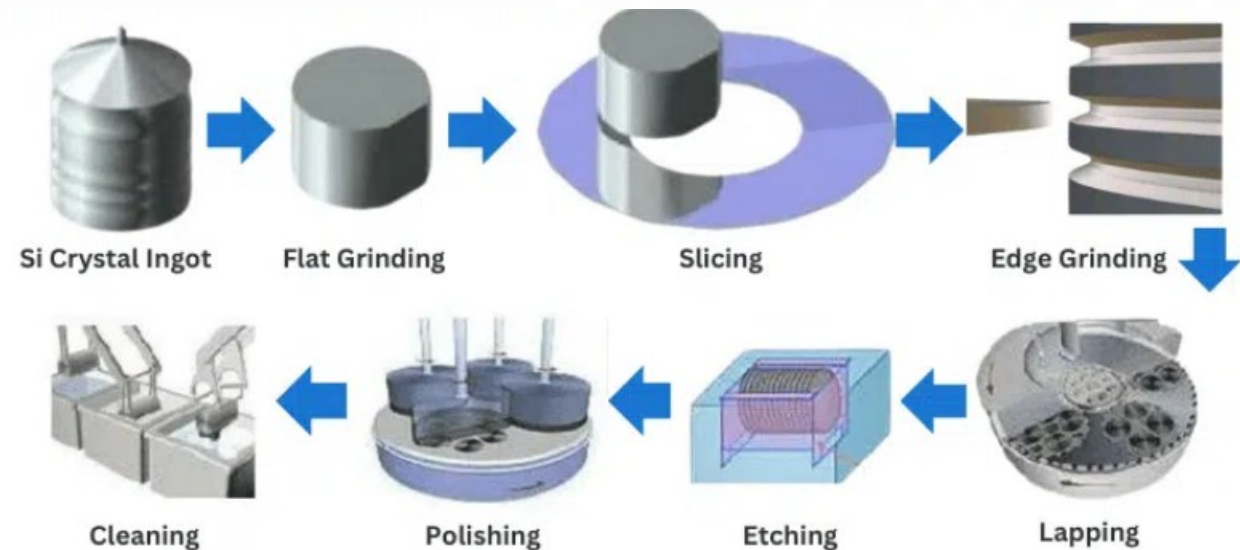
SILICON WAFER



Bulk Silicon Wafer

Doping means the introduction of impurities into a semiconductor crystal to the defined modification of conductivity. Two of the most important materials silicon can be doped with, are boron (3 valence electrons = 3-valent) and phosphorus (5 valence electrons = 5-valent). Other materials are aluminum, indium (3-valent) and arsenic, antimony (5-valent).

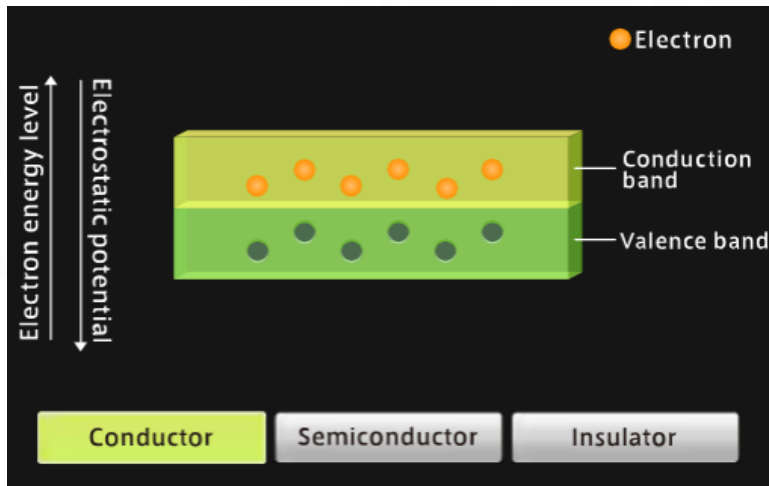
The dopant is integrated into the lattice structure of the semiconductor crystal, the number of outer electrons define the type of doping. Elements **with 3 valence electrons are used for p-type doping, 5-valued elements for n-doping**. The conductivity of a deliberately contaminated silicon crystal can be increased by a factor of 10^6 . **Silicon consists of four valence electrons!!!**



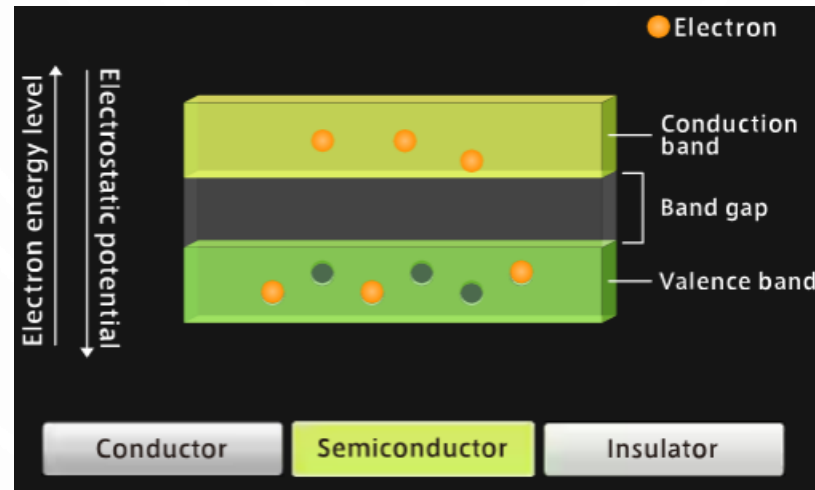
Silicon Wafer Manufacturing Process

SEMICONDUCTORS

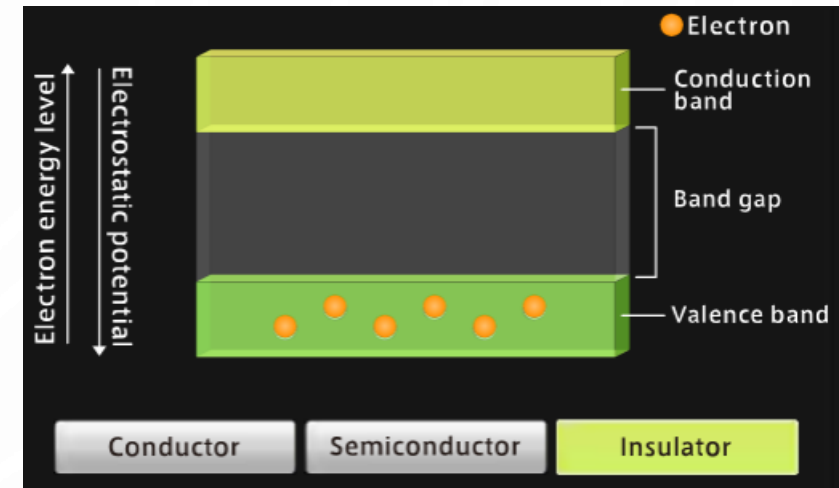
A semiconductor is a material that lets electrons flow through it under certain conditions. There are also materials, called conductors, that allow electrons to flow freely, and materials, called insulators, that prevent the flow of electrons.



conductor



semiconductor



insulator

Materials can be divided into three categories: conductors that allow electrons to flow through them, insulators that prevent the flow of electrons, and semiconductors that only let electrons flow under certain conditions. The difference between them may be best explained by the difference in their band gaps.

SEMICONDUCTORS

Materials can be divided into three categories: **conductors** that allow electrons to flow through them, **insulators** that prevent the flow of electrons, and **semiconductors** that only let electrons flow under certain conditions. The difference between them may be best explained by the difference in their band gaps.

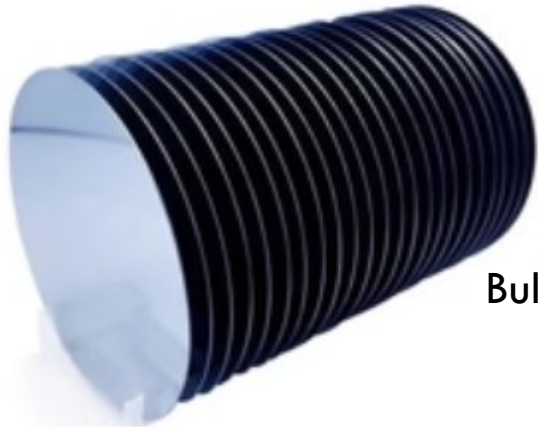
A band gap is an energy range in a material where no electron can exist. Conductors have no band gap, so electrons can freely move through them to generate an electric current. Metals including iron, copper, silver, gold, and aluminum are representative conductors. Insulators such as oil, glass, rubber, and ceramics have a large band gap which prevents the flow of electrons. Semiconductors, in contrast, have a small band gap, and **the flow of electrons and electron holes can be controlled by adding impurities to the material.**

Pure silicon and germanium crystals have insulator-like properties, and electricity hardly flows through them even when a voltage is applied. This is because their crystal lattice tightly keeps the electrons in place and hardly lets them move around.

When a very small amount of impurity such as phosphorus is introduced, however, it frees up some electrons and give the crystals conductor-like properties. Semiconductors containing impurities like phosphorus (P) and arsenic (As) that produce surplus electrons are called n-type semiconductors ("n" stands for negative), and those with impurities such as boron (B) and gallium (Ga) that create a deficiency of electrons are called p-type semiconductors ("p" stands for positive). In a p-type semiconductor, electron holes rather than electrons serve as charge carriers, behaving as if positively charged electrons are flowing.

When p-type and n-type semiconductors are joined, a so-called p-n junction is created.

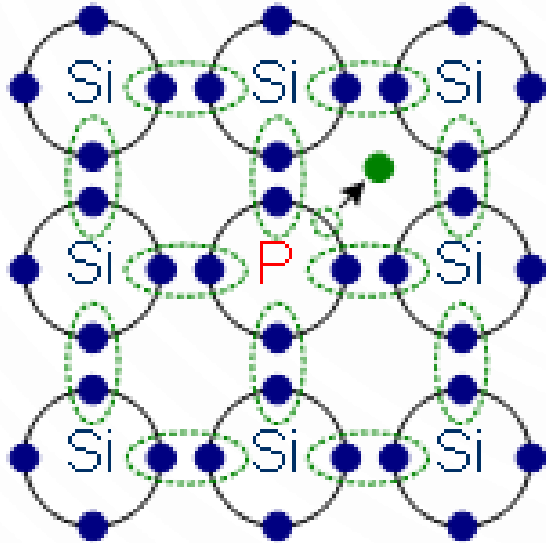
SILICON WAFER



Bulk Silicon Wafer

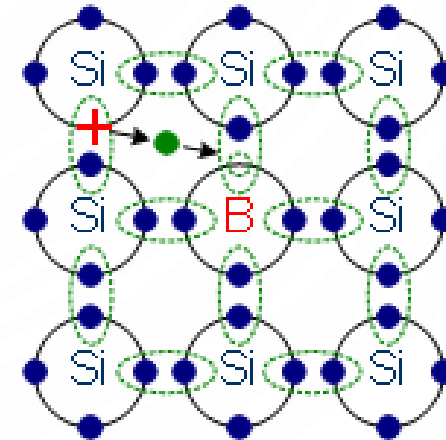
- Number of carriers is increased by introducing foreign atoms called impurities
- The process of introducing impurities is called doping
- Two Types of dopants: p type and n type
 - p-type dopants:** Boron (B), Gallium (G), Aluminum (Al)
 - n-type dopants:** Arsenics (Ar), Phosphorous (P), Antimony (Sb)

n-doping with phosphorus



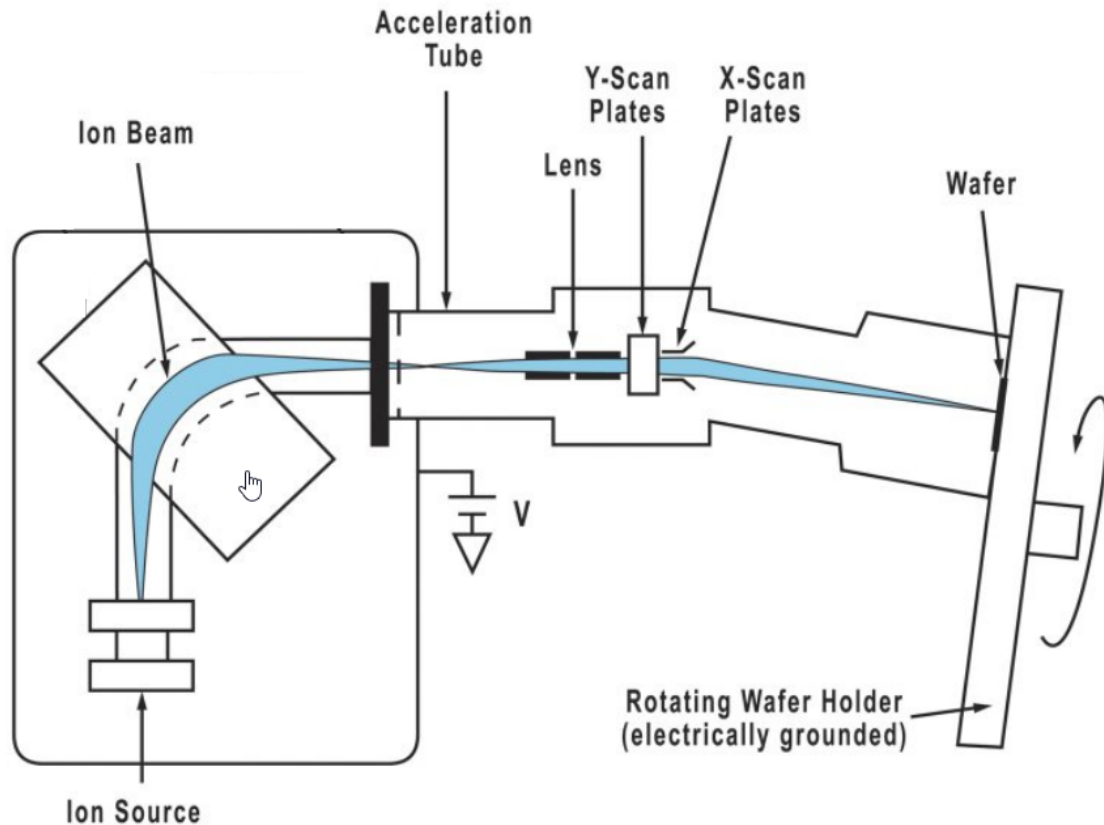
The phosphorus atom donates its fifth valence electron. It acts as a free charge carrier.

p-doping with boron



The free place on the boron atom is filled with an electron. Therefore a new hole („defect electron“) is generated. This holes move in the opposite direction to the electrons

SILICON WAFER



Doping of silicon wafer via ion beam

n-doping

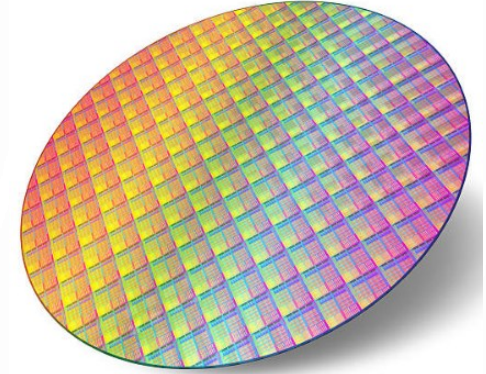
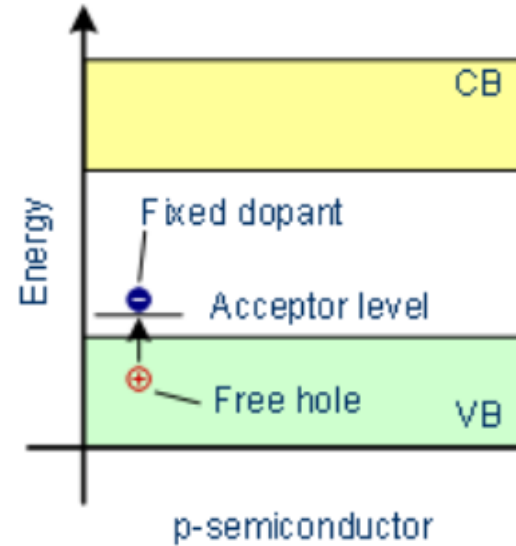
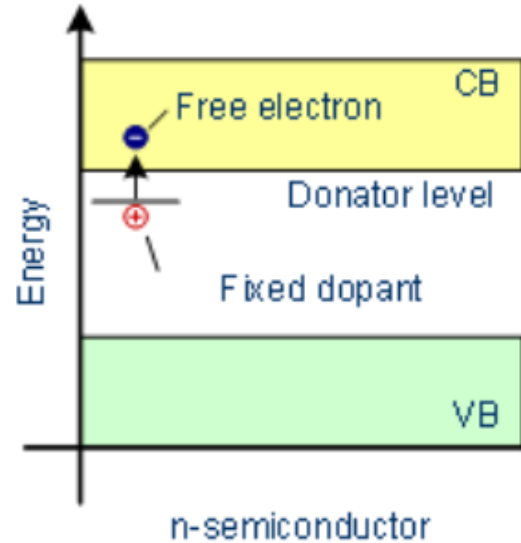
The 5-valent dopant has an outer electron more than the silicon atoms. Four outer electrons combine with every one silicon atom, while the fifth electron is free to move and serves as charge carrier. The dopant, which emits an electron, is known as an electron donor. Due to the higher number of free electrons those are also named as majority charge carriers

p-doping

In contrast to the free electron due to doping with phosphorus, the 3-valent dopant effect is exactly the opposite. The 3-valent dopants can catch an additional outer electron, thus leaving a hole in the valence band of silicon atoms. Therefore, the electrons in the valence band become mobile. With the inclusion of an electron, the dopant is negatively charged, such dopants are called acceptors. Due to positive holes these semiconductors are called p-conductive or p-doped. Analog to n-doped semiconductors, the holes are the majority charge carriers, free electrons are the minority charge carriers.

SILICON WAFER

Band model of doped semiconductors



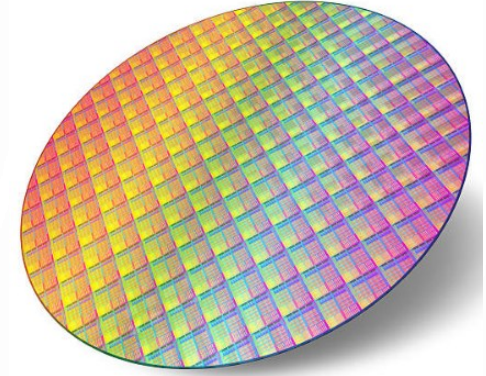
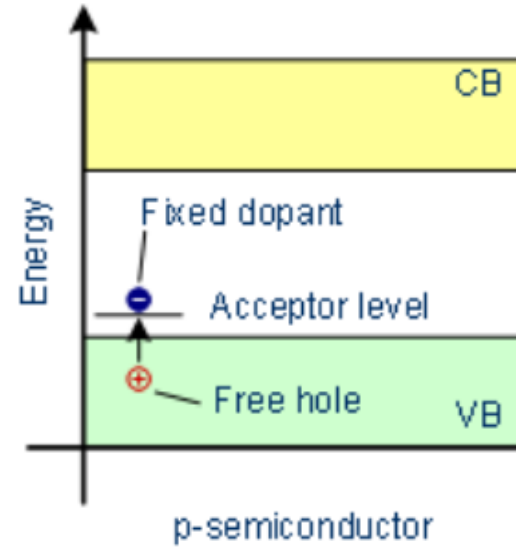
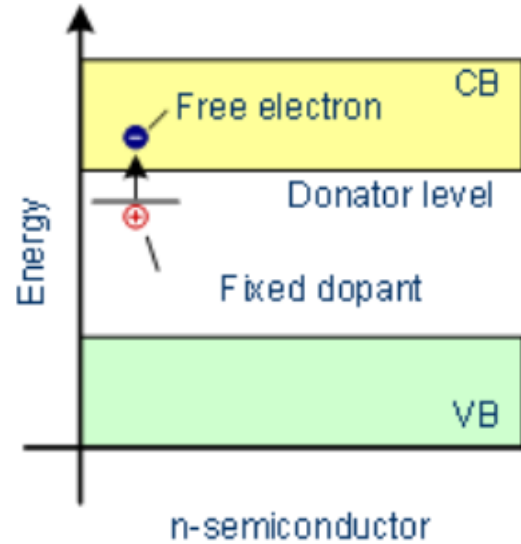
Silicon Wafer

By the introduction of a dopant with five outer electrons, in n-doped semiconductors there is an electron in the crystal which is not bound and therefore can be moved with relatively little energy into the conduction band. Thus, in n-doped semiconductors the donator energy level is close to the conduction band edge, the band gap to overcome is very small.

Analog, through introduction of a 3-valent dopant in a semiconductor, a hole is available, which may be already occupied at low-energy by an electron from the valence band of the silicon. For p-doped semiconductors the acceptor energy level is close the valence band.

SILICON WAFER

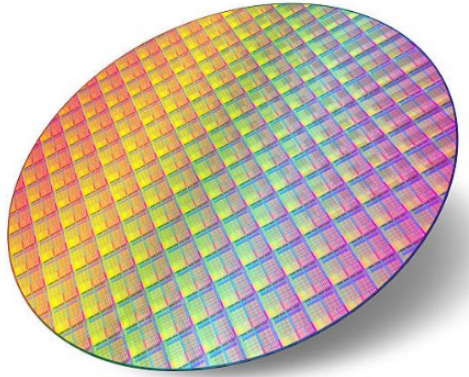
Band model of doped semiconductors



Silicon Wafer

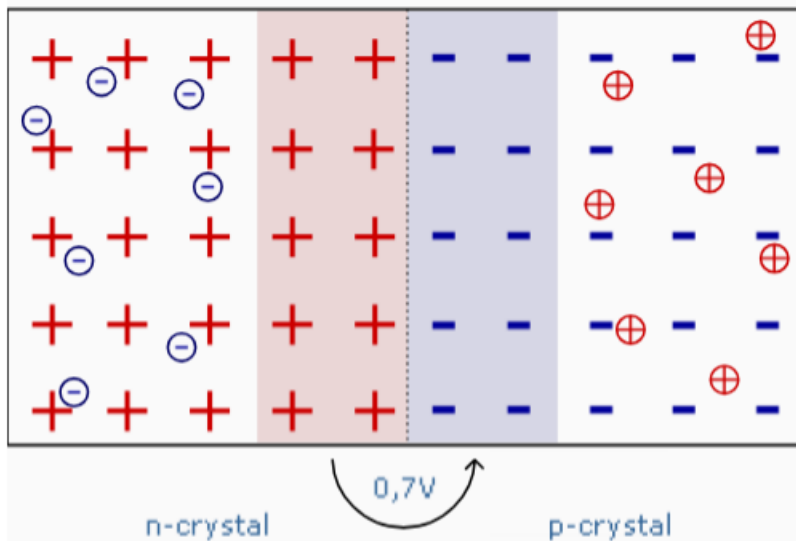
N- and p-doped semiconductors behave approximately equal in relation to the current flow. With increasing amount of dopants, the number of charge carriers increases in the semiconductor crystal. Here it requires only a very small amount of dopants. Weakly doped silicon crystals contain only 1 impurity per 1,000,000,000 silicon atoms, high doped semiconductors for example contain 1 foreign atom per 1,000 silicon atoms.

SILICON WAFER AND P-N JUNCTION



Silicon Wafer

The p-n junction is the transition area between two n- and p-doped semiconductor crystals. In this area there are no free charge carriers, since the free electrons of the n-conductor, and the holes of the p-doped crystal in the vicinity of the interface recombine with each other, which means that the electrons fill the holes. This charge movement (diffusion) is obtained in consequence of a concentration gradient: since there is only a few number of electrons in the p-area and only a few number of holes in the n-region, the majority charge carriers (electrons in the n-crystal, holes in the p-crystal) move into the contrary doped semiconductor.



1. Free moveable charge carriers at the interface of the n- and p-crystal recombine with each other. Thus a depletion zone without charge carriers remains.
2. Due to the lack of electrons a positive region remains in the n-crystal, as a negative zone is generated in the p-crystal. At a certain level of potential gradient the recombination of the electrons and holes comes to a standstill.

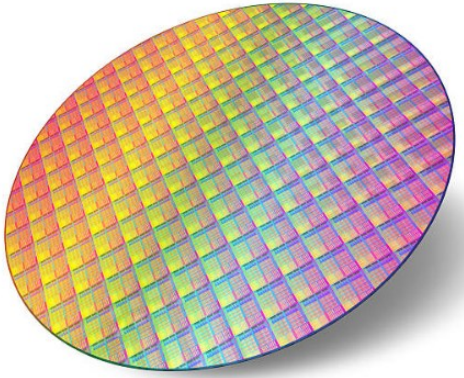
— fixed negative charge
(boron ion)

+ fixed positive charge
(phosphorus ion)

⊖ negative charge carrier (electron)

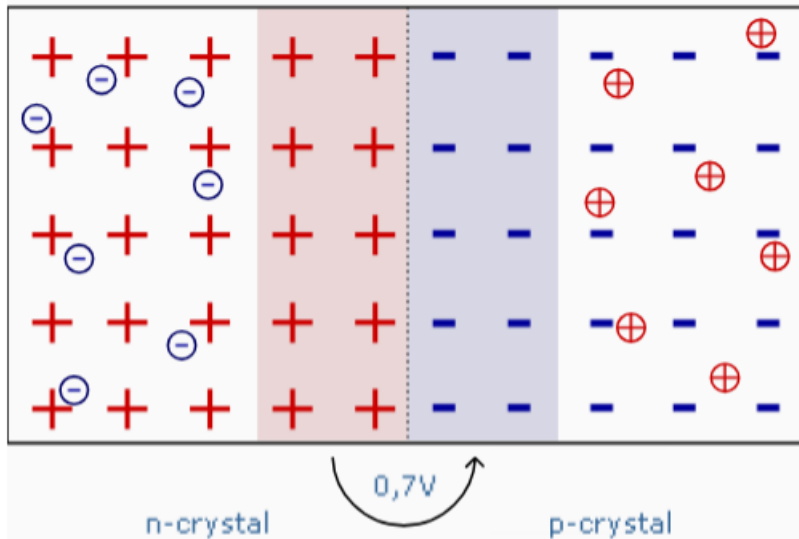
⊕ positive charge carrier (hole)

SILICON WAFER AND P-N JUNCTION



Silicon Wafer

The regions near the interface are loaded due to the loss of free charge carriers (positive charge in the n-crystal, negative charge in the p-crystal). The more charge carriers recombine, the greater the depletion zone and thus the voltage difference of n- and p-crystal. With a certain amount of this potential gap, the recombination of holes and electrons comes to a complete standstill, the charge carriers can no longer overcome the electric field. In silicon this limit is at about 0.7 V.



1. Free moveable charge carriers at the interface of the n- and p-crystal recombine with each other. Thus a depletion zone without charge carriers remains.
2. Due to the lack of electrons a positive region remains in the n-crystal, as a negative zone is generated in the p-crystal. At a certain level of potential gradient the recombination of the electrons and holes comes to a standstill.

+ Donors

- Acceptors

— fixed negative charge
(boron ion)

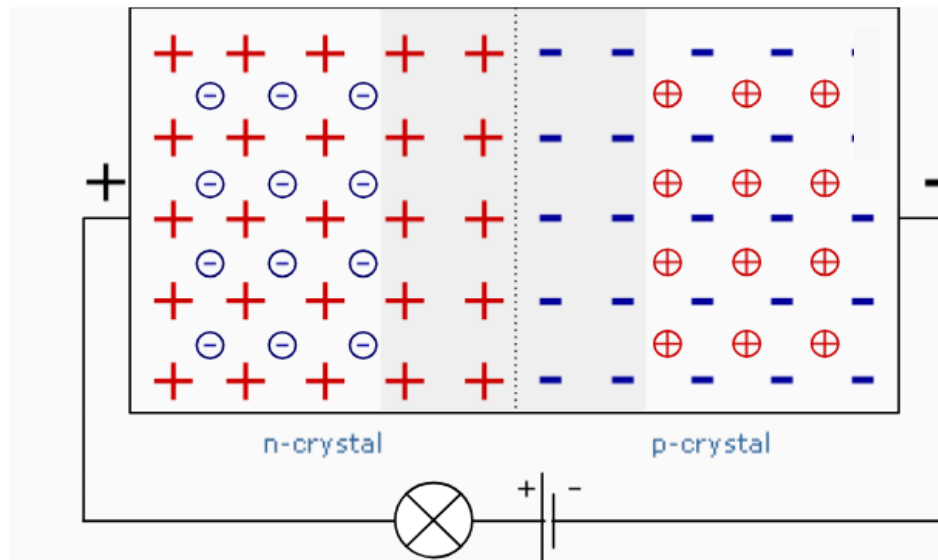
+ fixed positive charge
(phosphorus ion)

⊖ negative charge carrier (electron)

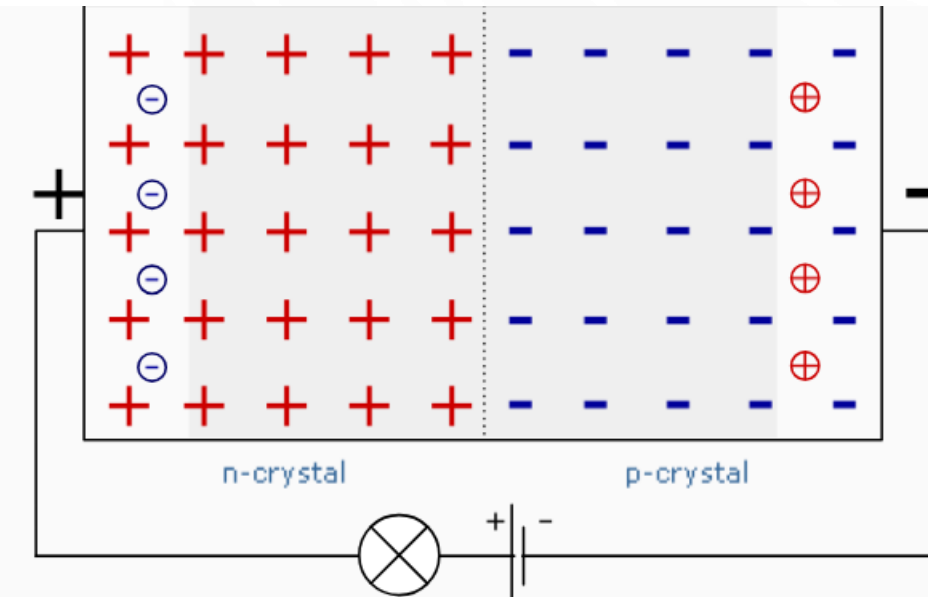
⊕ positive charge carrier (hole)

SILICON WAFER AND P-N JUNCTION

p-n junction with an external applied voltage

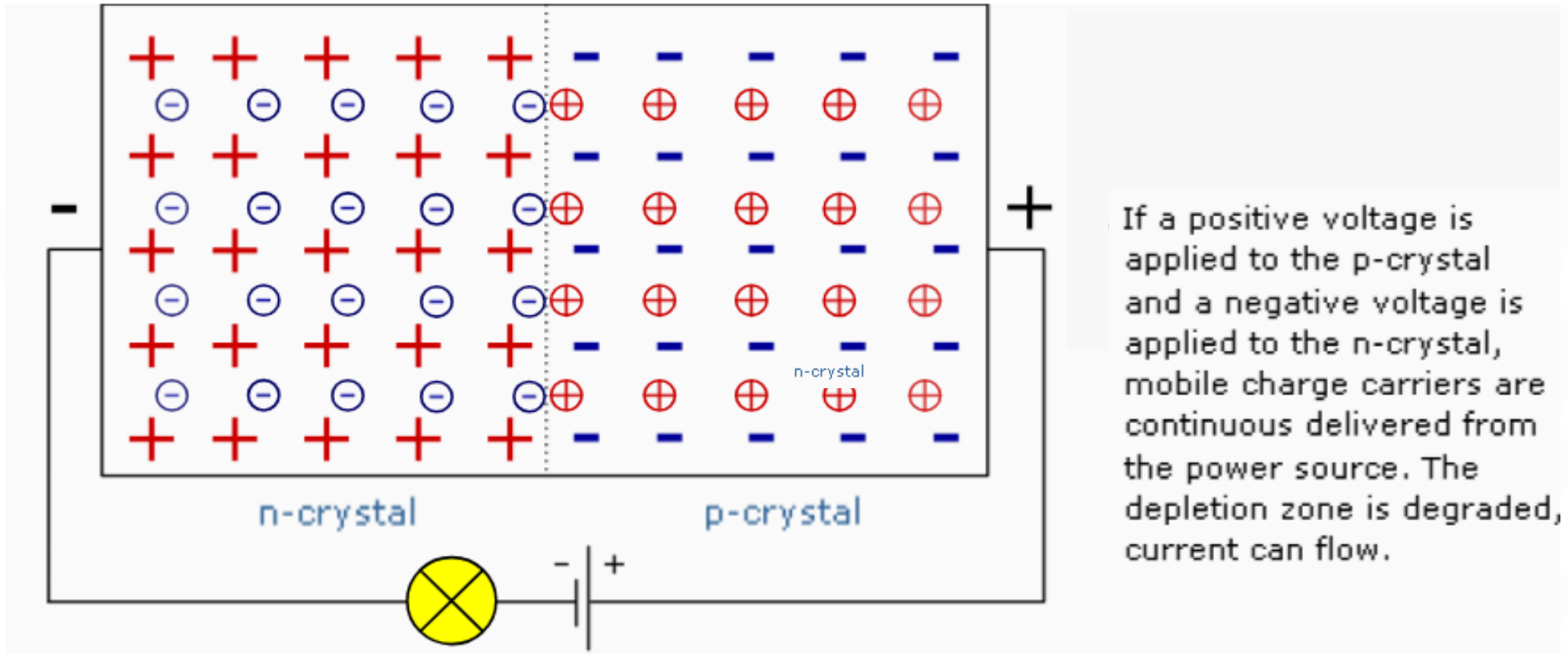


1. Without an applied voltage charge carriers at the interface can recombine. A depletion zone remains.



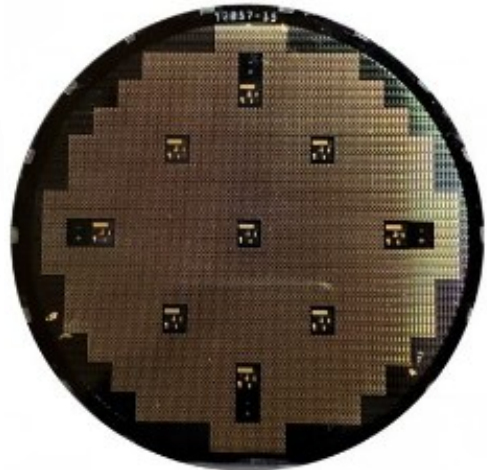
If a positive voltage is applied to the n-crystal and a negative voltage is applied to the p-crystal, the oppositely charged mobile charge carriers are attracted. The depletion zone increases, a current flow is not possible.

SILICON WAFER AND P-N JUNCTION

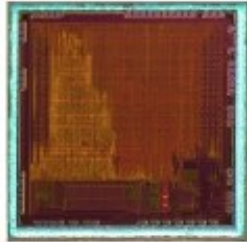


Mobile charge carriers move through the p-n junction, the depletion region becomes smaller

SILICON WAFER AND PACKAGING



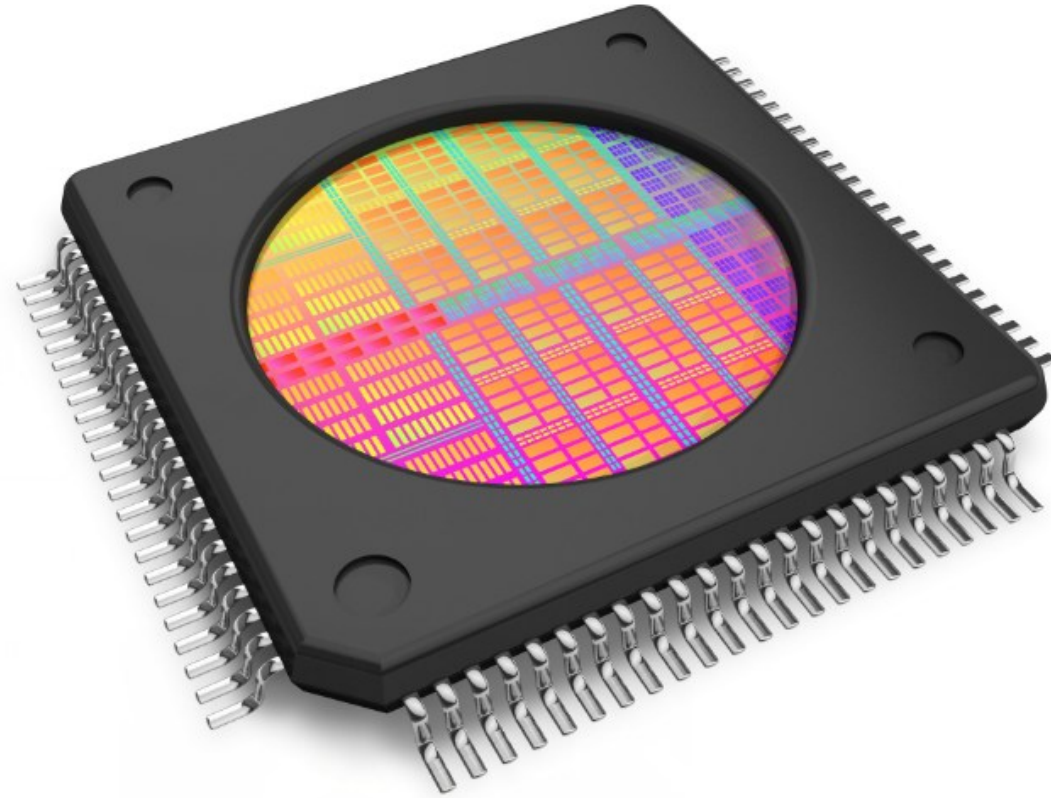
WAFER



SINGLE CHIP (DIE)



PACKAGED
SEMICONDUCTOR
DEVICE



PACKAGE TYPES

- used in Low&Medium Power Applications
- switching speed of transistors is much higher than the thyristors
- are extensively used in dc-dc and dc-ac converters
- can be turned “ON” and “OFF” by control signals

Surface Mount Device
(SMD)



Through Hole Technology (THT)
e.g. TO 247



Module Package (high power applications)

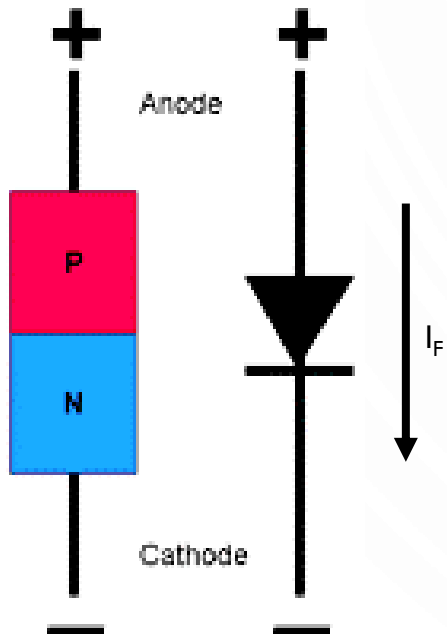


DIODES AND TRANSISTORS

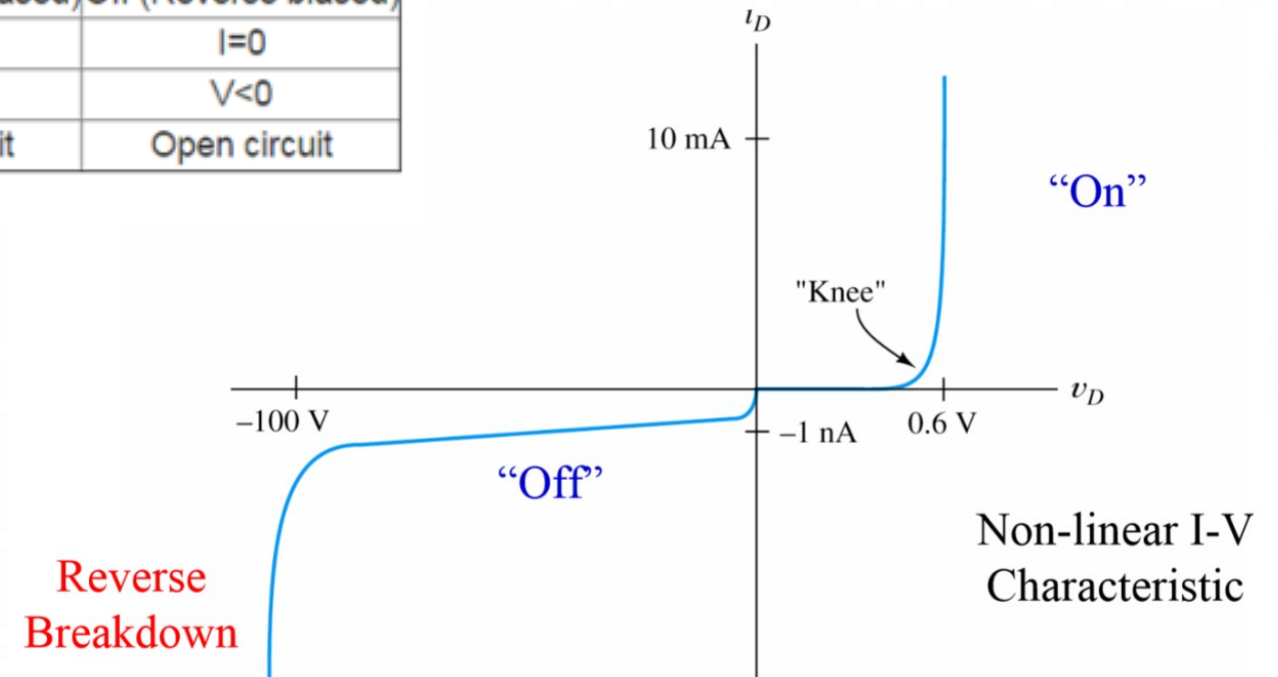
- Diode application examples
 - Rectifiers, ESD protection, voltage doublers, DCDC converters etc.
- Transistors
 - Switches, amplifiers etc.

DIODE

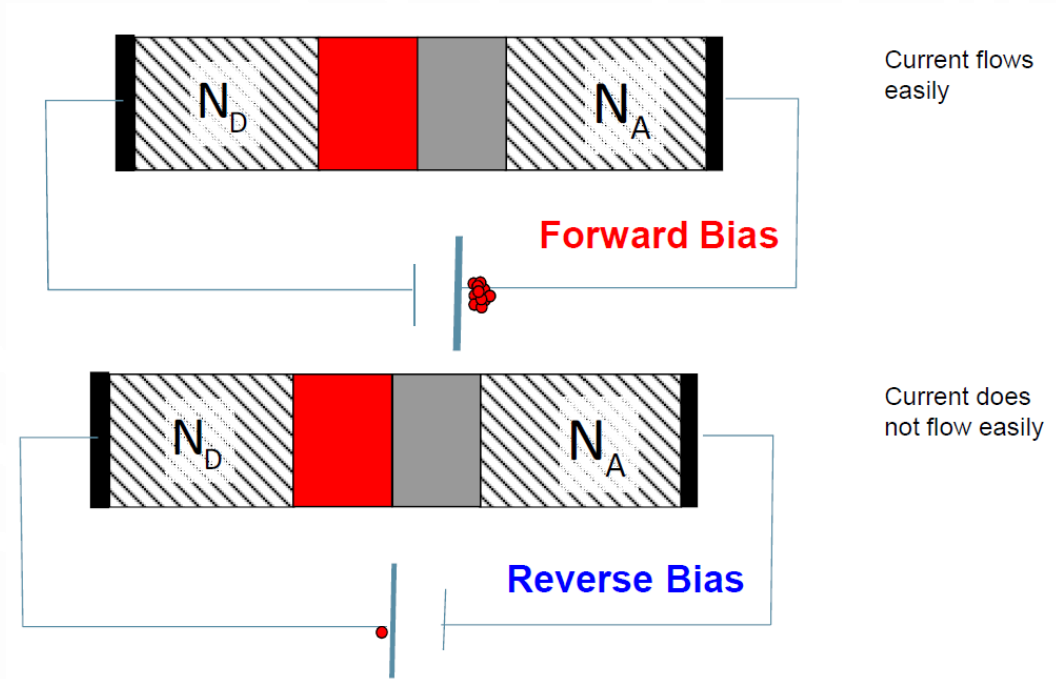
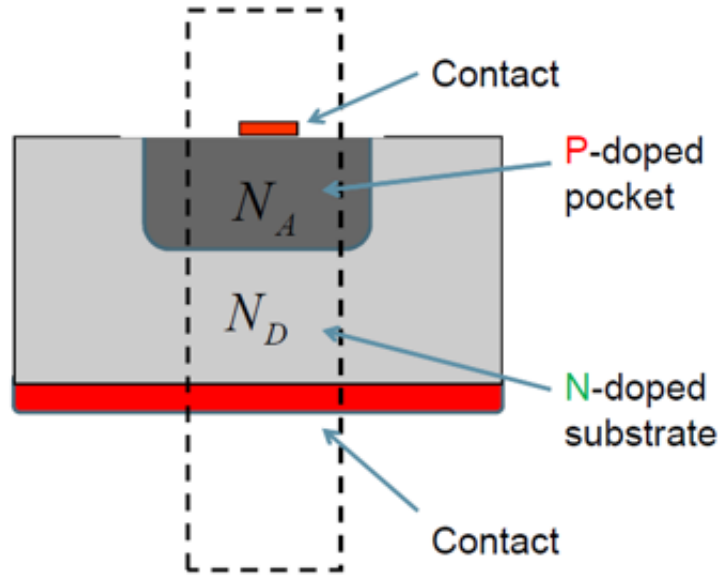
Diodes are made of silicon p-n junction with two terminals, anode and cathode



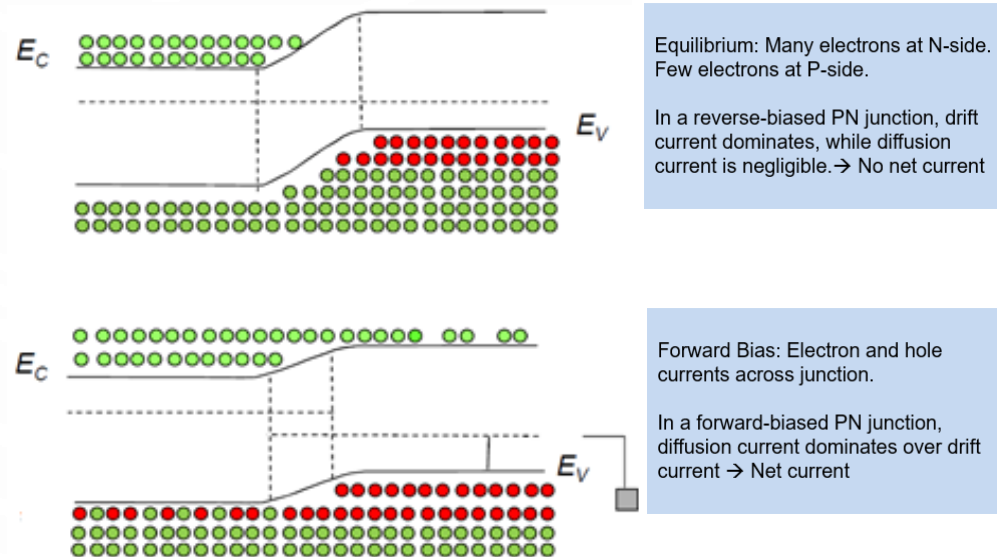
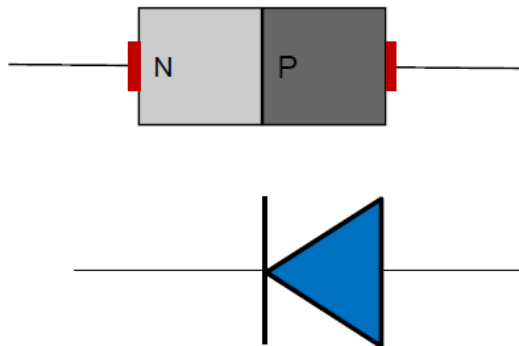
Ideal Diode Characteristics		
Operation Mode	On (Forward biased)	Off (Reverse biased)
Current Through	$I > 0$	$I = 0$
Voltage Across	$V > 0$	$V < 0$
Diode looks like	Short circuit	Open circuit



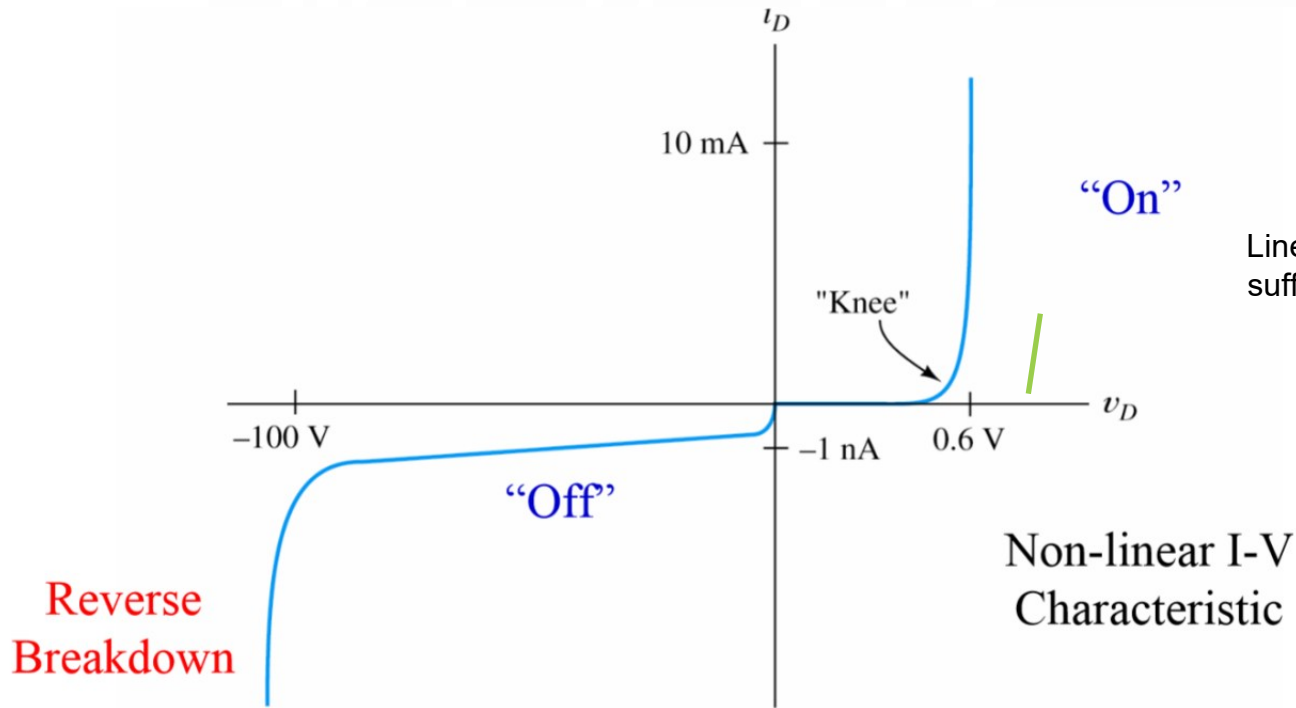
DIODE



Symbol

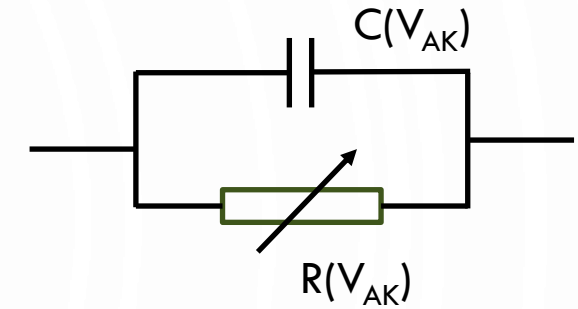


DIODE

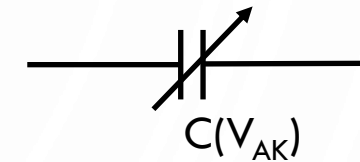


n = Emission coefficient ($1 < n < 2$)
 $k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant and
 $q = 1.60 \times 10^{-19}$ C is the magnitude of the electrical charge of an electron. At a temperature of 300 K, we have

$$V_T \cong 26 \text{ mV}$$



Small-signal equivalent circuit of forward-biased diode



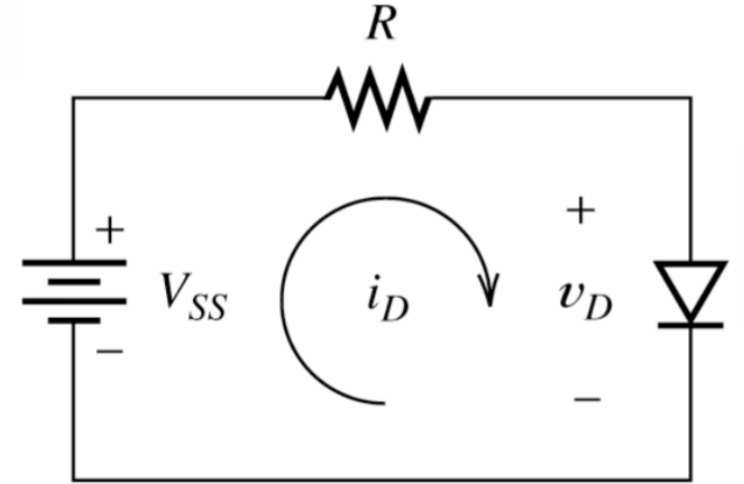
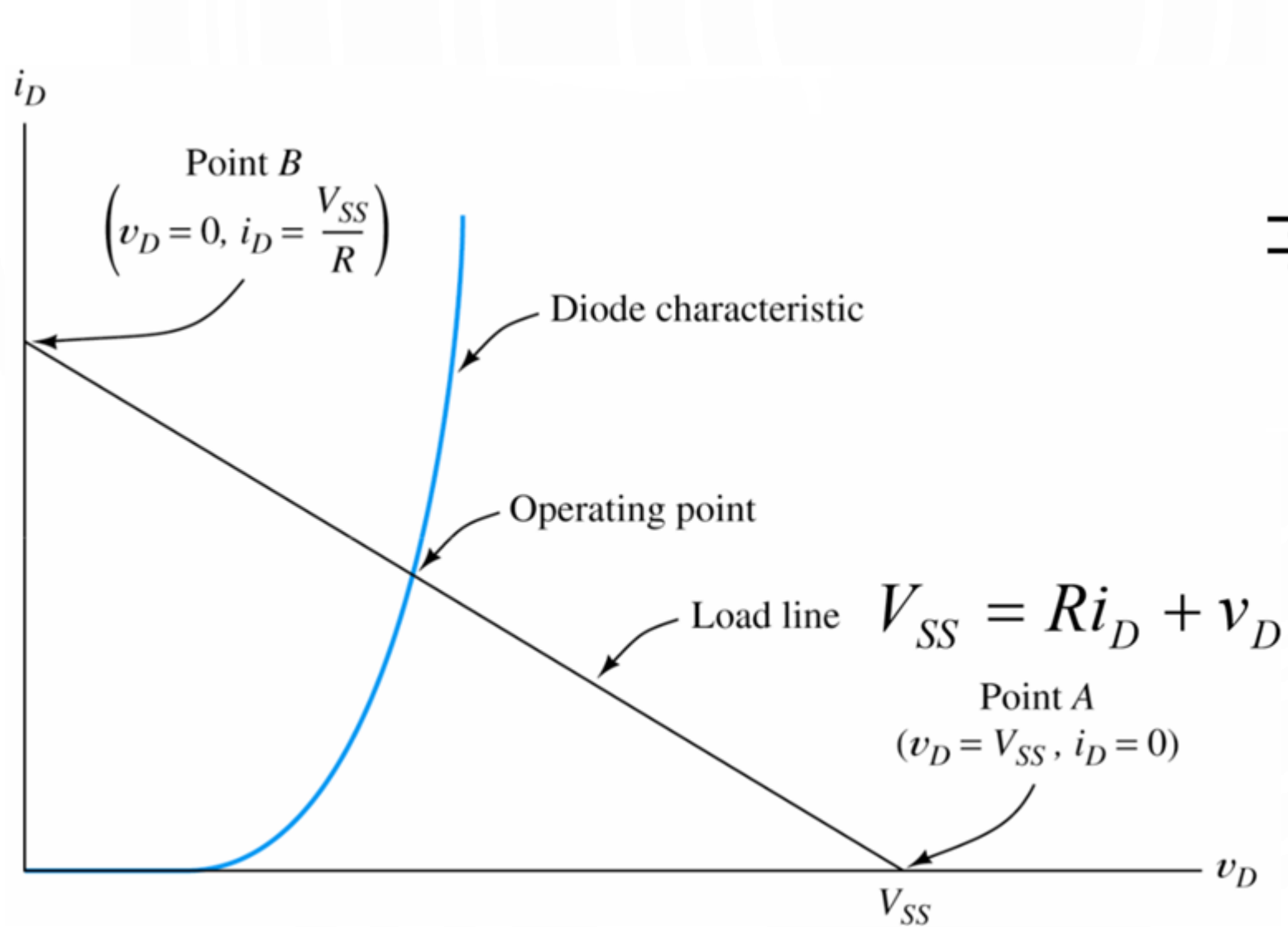
Small-signal equivalent circuit of reverse-biased diode

$$i_D = I_s \left[\exp\left(\frac{v_D}{nV_T}\right) - 1 \right] \quad V_T = \frac{kT}{q}$$

$$i_D \approx I_s \exp\left(\frac{v_D}{nV_T}\right) \text{ for } v_D \gg V_T$$

$i_D \approx -I_s$ for $v_D \ll -V_T$ until reaching reverse breakdown

DIODE



References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.